

Project Lab Level 5: Choose, Improve, and Present

Goal

Choose one course project, improve it from test evidence, prove its normal and failure routes, and present the result to the class in three clear minutes.

By the end of class, you can:

- choose one of your four projects for a specific person or job;
- use a real miss to choose one useful change;
- prove normal, failure, privacy, and stop routes separately;
- state the running cost and next safe step; and
- present the work without hiding the failure that improved it.

Start here

Bring proof from all four levels. Choose one project before class if possible. You may present live or use the supplied outage route. A saved run is honest proof when the service is down; pretending a recording is live is not.

Experienced builder? A deployed Vercel or Railway project must still show browser behavior, service health, and persistence as separate checks. A green deployment screen is not the whole proof.

Anchors / Ancias

- Show the work, the miss, the fix, and the switch. / Muestra el trabajo, el fallo, la corrección y el interruptor.
- Working is not the same as proven. / Funcionar no es lo mismo que estar comprobado.
- The audience should be able to repeat the check. / El público debe poder repetir la prueba.

Choose one project

Pick Documents, Watchman, Local Model, or Front Desk. Name one user and one job. Experienced builders may instead bring one of the built course-edition repositories from the project catalog, forked and changed, if the instructor agreed before class. Choose the project with the clearest useful proof, not the one with the most code.

Safety stop / Alto de seguridad

Clean the screen before presenting. Remove names, private records, browser history, tokens, environment values, notifications, and unrelated tabs. Use synthetic or public data. Never reveal a key to prove that a key exists.

Limpia la pantalla antes de presentar. Quita datos privados, claves, notificaciones y pestañas ajenas. Usa datos públicos o sintéticos.

Task 1: Build the proof board

Collect eight items:

1. Ownership: the files, account, or named local model are yours.
2. Before state: the miss, weak result, or unmet need.
3. Tested change: what you changed and why.
4. Normal run: one fresh input and expected result.
5. Failure route: missing data, bad input, attack, outage, or unavailable dependency.

6. Privacy or security fence: source receipt, local boundary, removed secret, or approval gate.

7. Stop or fallback: the switch used or the saved route opened.

8. Cost and reflection: fill the cost sheet with current provider price links, model and hosting assumptions, expected use, and one hour of human review; compute the cost floor; then name one next safe improvement. These are classroom estimates, not a CSR price.

Success check: a partner can locate all eight without hearing your speech.

Task 2: Run the final test

Write expected results first. Run one normal input and one failure input not used in the original level. Use the switch or fallback once. Record actual results. Fix only a blocking miss; do not rebuild the project during rehearsal.

For a deployed build, check separately:

- browser behavior from a fresh page;
- API or service health;
- saved state after a restart or new request, when persistence is claimed.

Task 3: Present to the class

Use the three-minute rubric:

- 30 seconds: who it helps and the outcome;
- 60 seconds: fresh normal run;
- 45 seconds: miss, change, and improved result;
- 30 seconds: fence plus stop or fallback;
- 15 seconds: cost and next safe step.

The class scores works, honest, and theirs from 1 to 3. Scores guide revision. They do not rank learners. Everyone receives applause.

Builder extension: show a Vercel frontend or Railway service only if the three separate checks above are visible and no production secret or private data is shown.

Quick check before proof

1. Can another person repeat the normal test?
2. Is the failure route visible?
3. Did you use the switch or fallback?
4. Is every screen safe for the room and recording?

Pass this level

Eight items: ownership, before state, tested change, normal run, failure route, privacy or security fence, stop or fallback, and cost plus reflection; followed by a three-minute class presentation.

Save the worksheet as PROJECT-LAB-05.md or keep the paper copy.

If something fails

- Service down: use the dated outage packet and explain the expected live check.

- Network slow: use screenshots or a short saved run, clearly labeled.
- Project breaks during rehearsal: show the break, read the log, and use the fallback.
- Project breaks on stage: state the expected result, show the last verified receipt, and continue.
- Learner does not want to speak: use a recorded or partner-read script while the learner controls the evidence.
- Class is large: use table rounds, then one representative demo per table and collect every proof board.

Resumen en español

Tarea 1: reúne ocho pruebas. Tarea 2: ejecuta una prueba normal, una falla y el interruptor o alternativa. Tarea 3: presenta durante tres minutos el resultado, el fallo, la corrección, el límite y el costo.

Words for Level 5

- Proof board: the eight artifacts supporting the project claim.
- Normal route: the expected use when dependencies work.
- Failure route: safe behavior when input or a dependency fails.
- Persistence: saved state that remains after a restart or later request.
- Rehearsal: a timed test of both the build and its explanation.
- Cost floor: the total expected monthly operating cost.